

Statistical Analysis Report: Income, Education, and Hours Worked

UCI Adult (Census Income) Dataset

Author: Timothy

Overview

This report analyzes the UCI Adult (Census Income) dataset to examine whether hours worked per week and education level are associated with whether a person earns more than \$50,000 per year. The dataset is publicly available from the UCI Machine Learning Repository at <https://archive.ics.uci.edu/dataset/2/adult>. The analysis was carried out in the accompanying notebook, `analysis.ipynb`, using descriptive statistics, three visual models, and a hypothesis test.

Dataset Description

The Adult dataset was extracted from the 1994 U.S. Census Bureau database and contains 32,561 rows and 15 columns describing individual survey respondents. Each row represents one person, with numeric variables including `age`, `education_num` (years of education encoded numerically), `capital_gain`, `capital_loss`, `hours_per_week`, and `fnlwgt` (a census sampling weight), alongside categorical variables including `workclass`, `education`, `marital_status`, `occupation`, `relationship`, `race`, `sex`, `native_country`, and `income` (the target variable, coded as `<=50K` or `>50K`). This report focuses primarily on `hours_per_week`, `education`, and `income`, which together support the analytical question of whether working hours and education level relate to income.

Methods

Descriptive statistics (`df.describe()` for numeric columns and `value_counts()` for categorical columns) were computed first, following the general principle that any downstream statistical analysis should begin with an initial, structured look at the data's distributions, missing values, and coding before modeling or testing it (Lusa et al., 2024). This initial pass revealed that `hours_per_week` is concentrated near the standard 40-hour work week and that `income` is imbalanced (about 76% `<=50K`), both of which shaped the choice of a group-comparison test rather than, for example, a simple correlation.

Three visual models were chosen to represent different aspects of the data: a boxplot (Figure 1) to directly compare the distribution of hours_per_week between the two income groups, since boxplots are well suited to comparing a numeric variable's median and spread across a small number of categories; a bar chart (Figure 2) to show how the proportion of high earners changes across the ordered categories of education, since bar charts are appropriate for comparing a summary rate across categories; and a correlation heatmap (Figure 3) to check, in one view, whether any of the numeric variables were strongly related to each other in a way that might confound the hours/income comparison.

A two-sample Welch's t-test was chosen to test whether mean hours_per_week differs between the $\leq 50K$ and $> 50K$ income groups. Welch's version (unequal variances) was used instead of the standard Student's t-test because the two groups have very different sample sizes (24,720 vs 7,841) and, as Figure 1 shows, different spreads; Welch's t-test does not assume equal variances and is generally recommended over the pooled-variance t-test whenever group sizes or variances differ, since it maintains a more accurate false positive rate under those conditions. The t-test itself assumes that the sampling distribution of each group's mean is approximately normal, which is a reasonable assumption here given the very large sample sizes (n in the tens of thousands) and the Central Limit Theorem, even though the raw hours_per_week variable itself is not perfectly normally distributed.

Results

Figure 1 shows that the $> 50K$ group has a visibly higher median and a box shifted upward compared to the $\leq 50K$ group. Consistent with this, the mean hours_per_week was 38.84 for the $\leq 50K$ group and 45.47 for the $> 50K$ group. The Welch's t-test produced $t = 45.12$ with an associated p-value effectively at the limit of floating-point precision (reported by the software as 0.0, i.e., $p < .001$), so the null hypothesis of equal means was rejected at the 0.05 significance level. Figure 2 showed that the proportion of high earners rises from 0% among respondents with only a Preschool education to roughly 41.5% for Bachelor's degree holders and 73-74% for Prof-school and Doctorate holders, a far larger swing than any difference explained by hours worked alone. Figure 3 showed that none of the numeric variables were strongly correlated with each other (all pairwise correlations below 0.16 in absolute value involving hours_per_week), which suggests the hours/income relationship is not simply a byproduct of some other correlated numeric variable.

Interpretation for a Non-Technical Audience

In plain terms: people who reported earning more than \$50,000 a year tended to work noticeably more hours per week on average (about 45.5 hours) than people earning \$50,000 or less (about 38.8 hours), and this difference is very unlikely to be due to random chance alone. At the same time, education level shows an even bigger split -- workers with an advanced degree were far more likely to be high earners than workers with only a high-school education or less, regardless of how many hours they worked. In other words, working more hours is associated with higher income in this dataset, but how much education someone has is associated with an even larger difference in the chance of being a high earner.

Limitations and Potential Bias

One clear dataset limitation is that the data were collected in 1994 from U.S. Census records, so the findings describe labor-market patterns from three decades ago and may not reflect current wages, work patterns, or the cost-of-living context behind the \$50,000 income threshold. A second limitation is that several categorical columns (workclass, occupation, and native_country) contain unreported values coded as '?', which were not imputed or removed in this analysis and could bias results if missingness is not random -- for example, if certain occupations are systematically under-reported.

A documented source of potential bias in this specific dataset is that the \$50,000 income threshold used to define the target variable was chosen somewhat arbitrarily, and the dataset overrepresents some demographic groups relative to the U.S. population, which can propagate historical social and economic biases (for example, by sex or race) into any conclusions drawn from it (Ding et al., 2021). Because this report groups people only by income and education, it does not examine whether the hours/income relationship holds equally across sex, race, or other demographic groups, so the conclusions should not be read as applying uniformly to every subgroup in the data.

Reproducibility

Anyone can rerun this analysis by installing the pinned dependencies in requirements.txt with 'pip install -r requirements.txt' and executing analysis.ipynb top to bottom (for example, with 'python -m nbconvert --to notebook --execute analysis.ipynb'), using the included adult_income.csv file, which is the raw UCI adult.data file with an added header row. Recording the initial data-checking steps directly in the notebook, before any modeling or testing, is itself a documented practice for building a reproducible analysis on top of a solid foundation (Lusa et al., 2024).

References

Ding, F., Hardt, M., Miller, J., & Schmidt, L. (2021). Retiring Adult: New datasets for fair machine learning. *Advances in Neural Information Processing Systems*, 34, 6478-6490.

Lusa, L., Proust-Lima, C., Schmidt, C. O., Lee, K. J., le Cessie, S., Baillie, M., Lawrence, F., & Huebner, M. (2024). Initial data analysis for longitudinal studies to build a solid foundation for reproducible analysis. *PLOS ONE*, 19(5), e0295726.
<https://doi.org/10.1371/journal.pone.0295726>